

Bayesian Credit Risk

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This project applies a Bayesian variable selection approach, Stochastic Search Variable Selection (SSVS), to a logistic regression model for loan default prediction: a spike-and-slab prior on each coefficient yields a posterior inclusion probability (PIP) for every candidate predictor, allowing us to identify the most relevant risk factors, quantify their effect with posterior uncertainty, and use the resulting model to predict default probability for new borrowers.

1 Data

1.1 Dataset Description

The dataset used is the *Credit Risk Dataset*, published on Kaggle. It consists of $n = 32,581$ loan applications, each described by 11 candidate predictors and a binary target variable, `loan_status` (1 = default, 0 = no default). The predictors cover applicant demographics (`person_age`, `person_income`, `person_home_ownership`, `person_emp_length`), loan characteristics (`loan_intent`, `loan_grade`, `loan_amnt`, `loan_int_rate`, `loan_percent_income`), and credit history (`cb_person_default_on_file`, `cb_person_cred_hist_length`).

1.2 Data Preparation

```
library(tidyverse)
library(tidymodels)

df <- read.csv("credit_risk_dataset.csv")
```

Applicants with implausible age or employment length are removed, together with observations containing missing values.

```
df_clean <- df %>%
  filter(person_age < 100,
         person_emp_length < 100 | is.na(person_emp_length)) %>%
  drop_na()
```

This reduces the sample from $n = 32,581$ to $n = 28,632$ observations (3,949 removed, 12.1%). *Posterior Note:* `drop_na()` was applied to all raw columns rather than only to the retained variables but the impact on the final estimates is minimal given the large sample size.

```
df <- df_clean %>%
  select(-loan_grade, -loan_int_rate, -cb_person_cred_hist_length) %>%
  mutate(log_income = log(person_income)) %>%
  select(-person_income) %>%
  mutate(person_home_ownership = case_when(
    person_home_ownership == "OTHER" ~ "OWN",
    TRUE ~ person_home_ownership
  ))

#Standardization
continuous_vars <- c("person_age", "person_emp_length", "loan_amnt",
                    "loan_percent_income", "log_income")
df[continuous_vars] <- scale(df[continuous_vars])
```

```
#Dummy coding
df <- df %>%
  mutate(
    person_home_ownership = relevel(factor(person_home_ownership), ref = "RENT"),
    loan_intent = relevel(factor(loan_intent), ref = "PERSONAL"),
    cb_person_default_on_file = relevel(factor(cb_person_default_on_file), ref = "N")
  )
```

three variables are excluded from the predictors: `loan_grade` and `loan_int_rate`, since both are assigned by the lender after assessing applicant risk and would introduce circularity if used and `cb_person_cred_hist_length`, due to its high collinearity with `person_age` ($r \approx 0.88$). The OTHER level of `person_home_ownership` (0.3% of obs) is merged into OWN while `person_income` is log-transformed to correct its strong right skew. All continuous predictors are standardized and finally all categorical predictors are dummy-coded.

2 Model

2.1 Design Matrix

X is the design matrix. The intercept β_0 is handled separately in the Nimble model, with its own prior $\beta_0 \sim \mathcal{N}(0, 10^2)$. Because `model.matrix` with `-1` still fully dummy-codes the first categorical variable in the formula, the reference level of `person_home_ownership` (RENT) is dropped manually.

```
X <- model.matrix(
  ~ person_age + person_emp_length + loan_amnt + loan_percent_income +
    log_income +
    person_home_ownership + loan_intent + cb_person_default_on_file - 1,
  data = df
)
X <- X[, colnames(X) != "person_home_ownershipRENT"]
Y <- df$loan_status
n <- nrow(X)
p <- ncol(X)
```

The resulting matrix has $n = 28,632$ observations and $p = 13$ candidate predictors.

Type	Variable
Continuous	<code>person_age</code> , <code>person_emp_length</code> , <code>loan_amnt</code> , <code>loan_percent_income</code> , <code>log_income</code>
Home ownership (ref: RENT)	<code>person_home_ownershipMORTGAGE</code> , <code>person_home_ownershipOWN</code>
Loan intent (ref: PERSONAL)	<code>loan_intentDEBTCONSOLIDATION</code> , <code>loan_intentEDUCATION</code> , <code>loan_intentHOMEIMPROVEMENT</code> , <code>loan_intentMEDICAL</code> , <code>loan_intentVENTURE</code>
Prior default (ref: N)	<code>cb_person_default_on_fileY</code>

Table 1: Candidate predictors

2.2 Spike-and-Slab Prior Specification

To each coefficient β_j , SSVS places a two-component Gaussian mixture prior:

$$\beta_j \mid \gamma_j \sim (1 - \gamma_j) \mathcal{N}(0, \tau^2) + \gamma_j \mathcal{N}(0, cc \cdot \tau^2), \quad \gamma_j \sim \text{Bernoulli}(\theta_j), \quad \theta_j \sim \text{Uniform}(0, 1), \quad (1)$$

where $\gamma_j = 0$ places β_j in the “spike” component and $\gamma_j = 1$ places it in the “slab” component. Each γ_j has its own inclusion probability (θ_j , Uniform(0,1)). The intercept β_0 is given fixed prior and is not subject to selection.

The spike variance and the slab scale factor are set using the intersection-point method that yields $\tau^2 \approx 2.71 \times 10^{-4}$ (spike variance) and $cc \cdot \tau^2 \approx 2.71$ (slab variance).

```
library(nimble)

c_ss      <- 100
intersect <- 0.05
tau_ss    <- intersect / sqrt(2*log(c_ss)*c_ss^2/(c_ss^2-1))
tau2      <- tau_ss^2      # spike variance
cc        <- c_ss^2        # slab variance = cc * tau2

data      <- list(Y = Y)
constants <- list(n = n, p = p, X = X, tau2 = tau2, cc = cc)
```

2.3 Model Specification

Combining the likelihood with the intercept and priors, the Nimble model is:

```
code_logit <- nimbleCode({
  for (i in 1:n) {
    Y[i] ~ dbern(eta[i])
    logit(eta[i]) <- beta0 + inprod(X[i, 1:p], beta[1:p])
  }

  beta0 ~ dnorm(0, sd = 10)

  for (j in 1:p){
    sig2[j] <- equals(gamma[j], 0) * var_spike + equals(gamma[j], 1) * var_slab
    beta[j] ~ dnorm(0, sd = sqrt(sig2[j]))
    gamma[j] ~ dbern(theta[j])
    theta[j] ~ dunif(0, 1)
  }

  var_spike <- tau2
  var_slab  <- cc * tau2
})
```

The parameters monitored are β_0 , $\beta_{1:p}$, and $\gamma_{1:p}$

```
params <- c("beta0", "beta", "gamma")
```

Chains are initialized with β_0 centered at the logit of the observed default rate (≈ -1.28) rather than at zero, to speed up burn-in.

```
p0_default <- mean(Y)
logit_p0    <- log(p0_default / (1 - p0_default))

inits_fun <- function() {
  list(
    beta0 = rnorm(1, logit_p0, 0.1),
    beta  = rnorm(p, 0, 0.1),
    gamma = rbinom(p, 1, 0.5),
    theta = runif(p, 0, 1)
  )
}
inits <- list(inits_fun(), inits_fun(), inits_fun())
```

2.4 MCMC Configuration

The model is fit on the dataset with 3 chains of 80,000 iterations each, a burn-in of 10,000 iterations, and a thinning interval of 20, yielding 3,500 retained draws per chain.

```
niter <- 80000
burn <- 10000
thin <- 20
nchains <- 3

model_logit <- nimbleModel(code_logit, data = data, inits = inits[[1]],
                           constants = constants)
compiled_model <- compileNimble(model_logit)

mcmc_conf <- configureMCMC(model_logit, monitors = params)
compiled_mcmc <- buildMCMC(mcmc_conf)
compiled_mcmc <- compileNimble(compiled_mcmc, project = model_logit)

samples <- runMCMC(compiled_mcmc,
                   niter = niter, nburnin = burn, thin = thin,
                   nchains = nchains, inits = inits)
```

Nimble automatically assigns a sampler to each parameter based on its conditional distribution `mcmc_conf$printSamplers()` was used to verify this.

2.5 Convergence Diagnostics

Convergence is assessed through the Gelman–Rubin statistic $\hat{R}$ across the 3 chains and by plotting traceplots and autocorrelation functions for each parameter.

```
MCMCsummary(samples, Rhat = TRUE)

par(mfrow = c(3, 2))
var_names <- colnames(X)
for (j in 1:p) {
  plot(samples$chain1[, j], type = "l", col = "#009E73",
        main = paste(var_names[j], "Traceplot"))
  lines(samples$chain2[, j], col = "#E69F00")
  lines(samples$chain3[, j], col = "skyblue")
  acf(samples$chain1[, j], main = paste(var_names[j], "ACF"))
}
```

Variable	Mean	SD	$\hat{R}$	n.eff
Intercept (β_0)	−1.586	0.048	1.00	3,237
person.age	−0.008	0.012	1.00	10,727
person.emp.length	−0.015	0.013	1.00	11,179
loan.amnt	0.306	0.043	1.00	3,070
loan.percent.income	0.594	0.038	1.00	3,383
log.income	−0.716	0.041	1.00	3,122
home.ownershipMORTGAGE	−0.792	0.039	1.00	10,249
home.ownershipOWN	−2.099	0.090	1.00	10,500
loan.intentDEBTCONSOLIDATION	0.525	0.058	1.00	4,287
loan.intentEDUCATION	−0.239	0.063	1.00	3,384
loan.intentHOMEIMPROVEMENT	0.671	0.065	1.00	4,954
loan.intentMEDICAL	0.358	0.057	1.00	4,120
loan.intentVENTURE	−0.411	0.063	1.00	4,488
cb.person.default.on.fileY	1.128	0.040	1.00	10,333

Table 2: Convergence diagnostics

All $\hat{R}$ values for the β 's are equal to 1.00, below the common threshold of 1.1, showing good convergence. Effective sample sizes range from about 3,000 to over 10,000 demonstrating low autocorrelation and reliable posterior estimates.

For the inclusion indicators γ_j , $\hat{R}$ is well defined and close to 1 for the three variables whose chains occasionally switch between $\gamma_j = 0$ and $\gamma_j = 1$ across iterations (`person_age`, `person_emp_length`, and `loan_intentEDUCATION`). While for the remaining ten variables, $\hat{R}$ and `n.eff` are undefined (NaN) because γ_j never changes value across any of the retained draws. This reflects an universal agreement that these variables belong in the model.

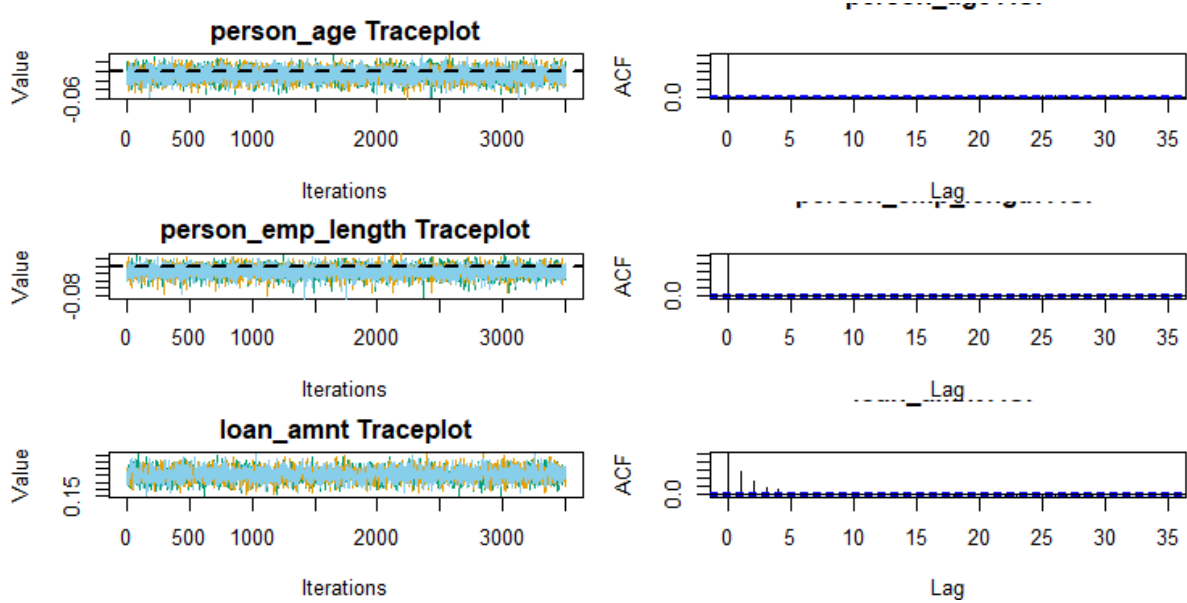


Figure 1: Traceplots (left) and ACF plots (right) for 3 representative variables

The three chains overlap closely in all three traceplots, with no visible trend or stickiness. The ACF for `person_age` and `person_emp_length` drops to zero almost immediately, consistent with their very high effective sample sizes; `loan_amnt` shows more persistent autocorrelation up to lag 5, consistent with its lower `n.eff`. Using Geweke's diagnostic convergence is further checked, comparing the mean of the first 10% and last 50% of each chain (under H_0 : equal means, giving a standard normal z -score)

```
library(coda)

geweke_table <- rbind(
  chain1 = geweke.diag(samples$chain1)$z,
  chain2 = geweke.diag(samples$chain2)$z,
  chain3 = geweke.diag(samples$chain3)$z
)

which(abs(geweke_table) > 1.96, arr.ind = TRUE)
```

across the 3 chains and the 27 monitored parameters, only 2 cells exceed $|z| > 1.96$: `beta[1]` (`person_age`) and `gamma[2]` (`person_emp_length`), both in chain 1. This is fewer than the ≈ 4 false positives expected by chance at the 5% level.

3 Variable Selection and Posterior Inference

3.1 Posterior Inclusion Probabilities and the Median Probability Model

The posterior inclusion probability (PIP) of each predictor is its posterior mean $E[\gamma_j \mid \text{data}]$, extracted from the MCMC summary. The Median Probability Model (MPM) selects every predictor with $\text{PIP} > 0.5$.

```
s <- MCMCsummary(samples, Rhat = TRUE)
col_gamma <- grep("gamma\\[", rownames(s))
post_mean_g <- s$mean[col_gamma]
names(post_mean_g) <- colnames(X)

par(mar = c(9, 4, 4, 1))
barplot(post_mean_g, names.arg = colnames(X), las = 2, cex.names = 0.6,
        main = "Posterior Inclusion Probabilities", col = "skyblue",
        ylab = "P(gamma = 1 | data)")
abline(h = 0.5, col = "blue", lty = 2, lwd = 2)

mdm_model <- post_mean_g > 0.5
mpm_selected <- colnames(X)[mdm_model]
```

Only two of the 13 candidate predictors fall below the 0.5 threshold, **person_age** (PIP = 0.016) and **person_emp_length** (PIP = 0.034), and are excluded from the MPM. The remaining 11 variables are included, with **loan_intentEDUCATION** the only one not at exactly 1 (PIP = 0.992).

Variable	PIP	MPM
person_age	0.016	Excluded
person_emp_length	0.034	Excluded
loan_amnt	1.000	Included
loan_percent_income	1.000	Included
log_income	1.000	Included
home_ownershipMORTGAGE	1.000	Included
home_ownershipOWN	1.000	Included
loan_intentDEBTCONSOLIDATION	1.000	Included
loan_intentEDUCATION	0.992	Included
loan_intentHOMEIMPROVEMENT	1.000	Included
loan_intentMEDICAL	1.000	Included
loan_intentVENTURE	1.000	Included
cb_person_default_on_fileY	1.000	Included

Table 3: Posterior inclusion probabilities

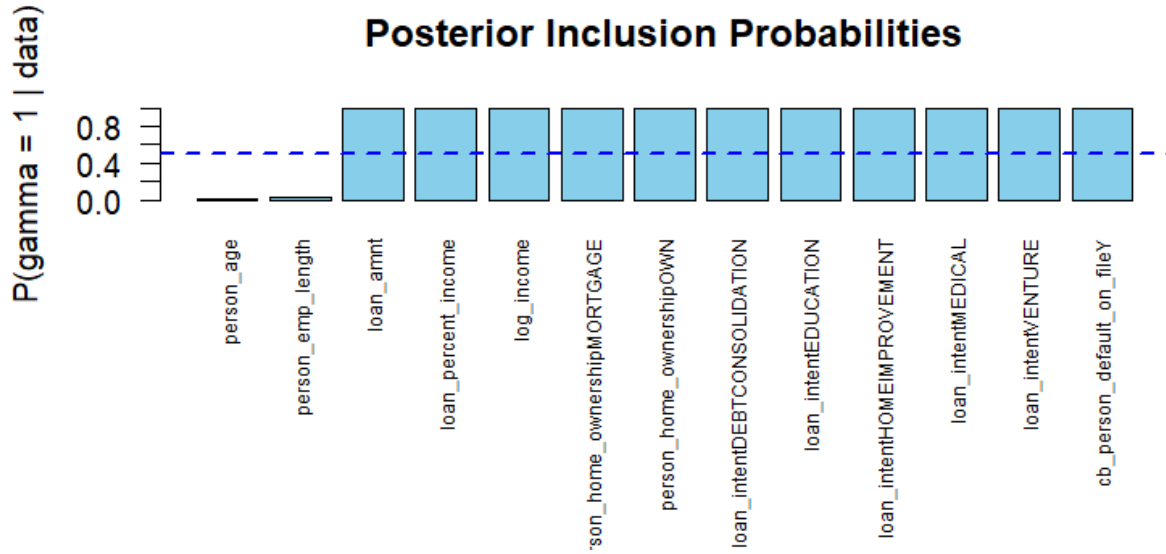


Figure 2: Posterior inclusion probabilities for all 13 candidate predictors, with the MPM threshold at 0.5.

3.2 Posterior Distributions of Selected Effects

The three chains are merged into a single sample matrix, and posterior draws for $\beta_{1:p}$ are extracted into a data frame. For each MPM-selected predictor, the posterior density is plotted together with a histogram of the pooled draws, the slab component of the prior (the density active when $\gamma_j = 1$), and the posterior median with 95% credible interval.

```
all_samples <- rbind(samples$chain1, samples$chain2, samples$chain3)
var_names <- colnames(X)

beta_cols <- paste0("beta[", 1:p, "]")
beta_samples <- as.data.frame(all_samples[, beta_cols])
colnames(beta_samples) <- var_names

par(mfrow = c(3, 4))
sd_slab <- sqrt(cc * tau2)

for (v in mpm_selected) {
  chain <- beta_samples[[v]]
  post_quant <- quantile(chain, prob = c(0.025, 0.5, 0.975))
  post_dens <- approxfun(density(chain, adj = 2))

  hist(chain, freq = FALSE, col = "gray", xlim = range(chain), main = v, xlab = "")
  lines(density(chain, adj = 2), col = "black", lwd = 3)
  curve(dnorm(x, 0, sd = sd_slab), from = range(chain)[1], to = range(chain)[2],
        add = TRUE, lty = 2, lwd = 2, col = "red")

  segments(post_quant[1], 0, post_quant[1], post_dens(post_quant[1]), col = "black",
           lty = 2, lwd = 2)
  segments(post_quant[3], 0, post_quant[3], post_dens(post_quant[3]), col = "black",
           lty = 2, lwd = 2)
  segments(post_quant[2], 0, post_quant[2], post_dens(post_quant[2]), col = "blue", lwd
           lty = 2)
}
```

All densities are approximately Normal, with well separated credible intervals that exclude zero; consistent with their posterior inclusion probabilities being at or near 1. The slab prior (red dashed curve) is essentially flat over the range covered by each posterior, reflecting a vague prior dominated by the data. `person_home_ownershipOWN` has by far the strongest protective effect (≈ -2.10), followed by `cb_person_default_on_fileY` (≈ 1.13), the strongest risk increasing

effect.

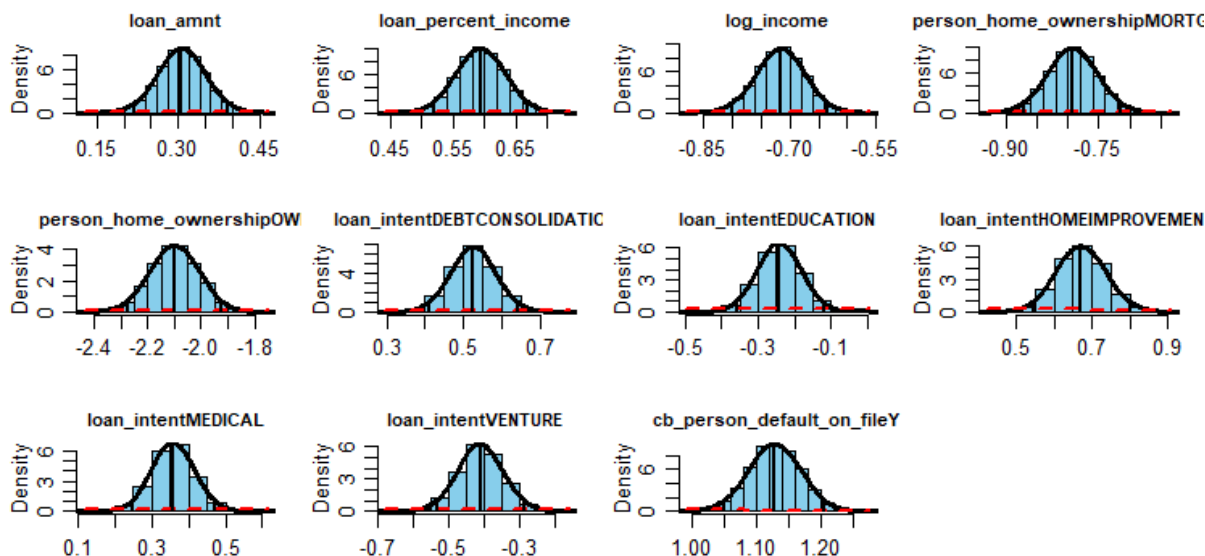


Figure 3: Posterior densities of the selected coefficients

3.3 Low-Risk vs High-Risk profiles

Posterior default probabilities are computed for two synthetic borrower profiles, constructed on the standardized covariate scale (0 corresponds to the sample mean and ± 1 to one standard deviation above/below it) with the remaining categorical predictors fixed.

- **Low-risk profile:** home owner (OWN), no prior default, personal loan (reference category), below-average loan-to-income ratio (-1 SD); age, income, loan amount, and employment length at the sample mean.
- **High-risk profile:** renter (RENT), prior default on file, debt-consolidation loan, above-average loan-to-income ratio ($+1$ SD); same average values for the remaining continuous predictors.

For each posterior draw $\beta^{(s)}$, the predicted default probability is obtained via the inverse logit of the linear predictor.

```
beta_cols <- paste0("beta[", 1:p, "]")
beta_post <- all_samples[, beta_cols]
beta0_post <- all_samples[, "beta0"]

profilo_basso <- c(person_age = 0, person_emp_length = 0, loan_amnt = 0,
  loan_percent_income = -1, log_income = 0,
  person_home_ownershipMORTGAGE = 0, person_home_ownershipOWN = 1,
  loan_intentDEBTCONSOLIDATION = 0, loan_intentEDUCATION = 0,
  loan_intentHOMEIMPROVEMENT = 0, loan_intentMEDICAL = 0,
  loan_intentVENTURE = 0, cb_person_default_on_fileY = 0)
profilo_basso <- profilo_basso[colnames(X)]

profilo_alto <- c(person_age = 0, person_emp_length = 0, loan_amnt = 0,
  loan_percent_income = 1, log_income = 0,
  person_home_ownershipMORTGAGE = 0, person_home_ownershipOWN = 0,
  loan_intentDEBTCONSOLIDATION = 1, loan_intentEDUCATION = 0,
  loan_intentHOMEIMPROVEMENT = 0, loan_intentMEDICAL = 0,
  loan_intentVENTURE = 0, cb_person_default_on_fileY = 1)
profilo_alto <- profilo_alto[colnames(X)]

prob_basso <- 1 / (1 + exp(-(beta0_post + beta_post %*% profilo_basso)))
```

```

prob_alto <- 1 / (1 + exp(-(beta0_post + beta_post %*% profilo_alto)))

hist(prob_basso, breaks = 40, freq = FALSE, col = rgb(0.2, 0.6, 0.9, 0.5), border = NA,
     xlim = range(c(prob_basso, prob_alto)), main = "P(default): low- vs high-risk
     ↪ profile")
hist(prob_alto, breaks = 40, freq = FALSE, col = rgb(0.9, 0.2, 0.2, 0.5), border = NA,
     ↪ add = TRUE)
abline(v = mean(prob_basso), col = "blue", lwd = 2)
abline(v = mean(prob_alto), col = "red", lwd = 2)

```

The posterior predictive default probability is 0.0137 for the low-risk profile, against 0.6595 for the high-risk profile. The two posterior distributions are non overlapping and individually narrow, reflecting the large sample size; the gap between the two profiles is driven jointly by home ownership, prior default history, loan intent, and loan-to-income ratio, the four strongest predictors.

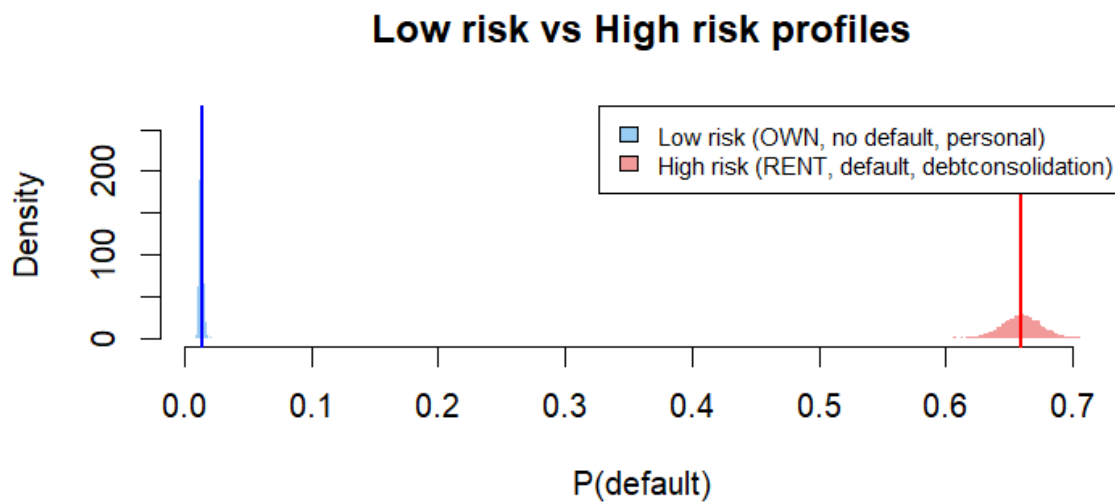


Figure 4: Posterior predictive distribution $P(\text{default})$ for the low-risk and high-risk profiles

4 Predictive Validation

Predictive performance is assessed with a frequentist logistic regression fitted only on the MPM selected predictors, on an 80/20 train/test split.

```
library(tidymodels)

df_model <- df
df_model$loan_status <- factor(df_model$loan_status)
levels(df_model$loan_status) <- c("NoDefault", "Default")

set.seed(123)
training_index <- sample(1:nrow(df_model), 0.8 * nrow(df_model), replace = FALSE)
df_train <- df_model %>% filter(row_number() %in% training_index)
df_test <- df_model %>% filter(!row_number() %in% training_index)

logreg_spec <- logistic_reg() %>%
  set_engine("glm") %>%
  set_mode("classification")

logreg_fit <- logreg_spec %>%
  fit(loan_status ~ loan_amnt + loan_percent_income + log_income +
      person_home_ownership + loan_intent + cb_person_default_on_file,
      data = df_train)

logreg_pred <- logreg_fit %>%
  predict(new_data = df_test, type = "prob") %>%
  bind_cols(actual = df_test$loan_status)

roc_data <- logreg_pred %>%
  roc_curve(truth = actual, .pred_Default, event_level = "second")
auc_val <- logreg_pred %>%
  roc_auc(truth = actual, .pred_Default, event_level = "second")

autoplot(roc_data) +
  labs(title = paste("ROC Curve - AUC =", round(auc_val$.estimate, 3)))
```

The model achieves an AUC of 0.806 on the test set. The ROC curve lies well above the diagonal reference line (random classifier) throughout the full range of thresholds. This confirms that the 11 MPM-selected predictors retain strong discriminative power for out-of-sample default prediction.

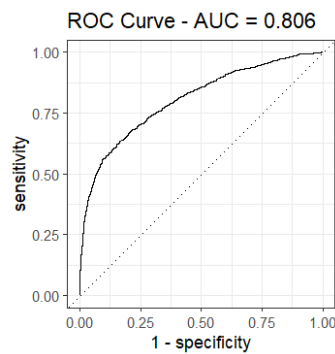


Figure 5: ROC curve